

Agent-based Simulation with Mobility Personas: An LLM-based Persona Labelling Approach for MATSim Synthetic Populations

Jingjun Li¹, Shiqi Sun¹, Francesco Ciari², Chengqi Lu³, Guy Walker¹, John Easton¹, David Flynn⁴, Philip Greening¹

- 1) School of Social Sciences, Heriot-Watt University, Edinburgh EH14 4AS, United Kingdom
- 2) Civil, Geological and Mining Engineering Department, Polytechnique Montreal, Montreal H3T0A3, Canada
- 3) Chair of Mobility System Planning, Technische Universität Dresden, 01069 Dresden, Germany
- 4) James Watt School of Engineering, University of Glasgow, Glasgow G12 8QQ, United Kingdom

E-mail: jingjun.li@hw.ac.uk

Introduction & Motivation

The urgent need for transport decarbonisation requires identifying low-cost transition pathways that account for social, technical, economic, and environmental challenges. However, the urban mobility system is, by its nature, a highly complex socio-technical system (Li et al., 2026), in which individual behaviours, transport services, infrastructure, and transport interventions dynamically interact. Agent-based modelling, therefore, provides a valuable means of representing these interactions and examining how heterogeneous individual decisions collectively produce system-level outcomes. Against this background, MATSim has become one of the most widely used agent-based transport simulation frameworks for modelling large-scale (typically urban-level) individual movements with different activity patterns over a single day. Typically, MATSim runs follow an iterative co-evolutionary process, in which agents execute daily plans, receive scores, and adapt their plans through replanning strategies such as rerouting, departure-time mutation, and mode changes.

In many applications, agents in MATSim remain behaviourally homogeneous at the level of mobility decision logic. Specifically, agents in MATSim may differ in activity scheduling, locations, mode availability, and network conditions, but their behavioural adaptations are often governed by a unified scoring function and replanning strategies. This becomes problematic when modelling mobility scenarios where responses to the same interventions may vary substantially across individuals. For example, agents facing similar physical and practical mobility constraints may still differ in their willingness to use public transport, perceptions of safety, environmental attitudes, or acceptance of emerging mobility services. These differences can lead them to respond to the same intervention in different ways. In principle, MATSim provides the flexibility to represent such behavioural heterogeneity by defining different subpopulations and assigning different scoring functions or replanning strategies. However, this raises a further methodological question: how should these subpopulations be defined in a way that is behaviourally meaningful to large synthetic populations?

To address this gap, we propose an approach that uses Large Language Models (LLMs) to label mobility personas across different synthetic populations in agent-based simulations. As a result, personas serve as an intermediate, aggregated behavioural layer between individual attributes and the MATSim simulation logic. Compared to conventional approaches such as rule-based statistical matching, LLM-based persona labelling reduces the reliance on costly primary data collection and ambiguously defined rules. This research, therefore, evaluates whether LLM-based persona labelling can provide a feasible prediction for synthetic populations with different profiles.

Methods

The starting point of this work is an existing cohort of transport user personas developed by the UK Department for Transport (DfT, 2023). In total, based on individual traits when using transport, the

population aged 16 and beyond in England is divided into 12 personas and 9 segments, namely: Less mobile, car reliant; Young urban families; Older, less affluent; Comfortable empty-nesters; Suburban families; Heavy car users, frequent flyers; Elderly and low income without cars; Urban professionals without cars; and Young low income without cars. A brief demonstration of the respective personas and segments is available at: <https://www.gov.uk/guidance/transport-user-personas-understanding-different-users-and-their-needs>.

The original DfT segmentation was developed from a transport choices segmentation survey that collected data on respondents' demographic characteristics, attitudes, behaviours, and car ownership-related factors. These survey variables were used in a cluster analysis to produce the nine segments, together with a reduced set of these drivers through statistical discriminant analysis (referred to as "golden questions"). In principle, this allows a new respondent to answer a small set of questions in around 5 minutes and then be allocated to one of the established transport user segments. However, the golden questions still depend substantially on responses related to attitudes, perceptions, and subjective behavioural variables, such as whether a person believes that successful people tend to travel by car rather than by bus. Therefore, it is difficult to apply directly at the scale of an entire urban population, and it is not directly applicable to synthetic populations whose agents do not exist as real respondents (Sun et al., 2026).

To test whether LLMs can bridge this gap, we use data from the 2022 DfT transport choices survey, in which the probability that a respondent belongs to each DfT segment is estimated using coefficients from the golden question models. Then, 50 respondent records are sampled from the survey dataset. For each respondent, the DfT segment allocation derived from the golden questions is treated as the reference classification. We then construct a reduced attribute profile using 20 attributes that are either available in the synthetic population data or can be proxied by it. At present, in addition to basic demographic information, the attributes also cover employment and education status, travel habits, health issues and household information. Then, we use an ensemble of LLMs to evaluate a reduced-information inference problem: if the golden questions are unavailable, can LLMs reconstruct the segmentation that the DfT model would have produced? For each individual, LLMs are asked to infer a probability distribution across nine segments, where the segment distributions are compared with the reference DfT golden questions model for validation. The LLMs are also instructed to produce a confidence level, so that the heterogeneity and accuracy of the predictions can be well balanced by a decision fusion model that calculates a weighted average of the LLMs' results (see Sun et al., 2026, for details on the decision fusion model). The experiment, therefore, provides an initial test of whether mobility personas can be inferred from synthetic-population-like attributes and subsequently attached to MATSim agents as an additional behavioural layer.

Outcomes

To assess prediction quality, we compare the segment rankings produced by each LLM and the decision fusion model with the reference segment assigned by the DfT model. The prediction quality is assessed by Top-k accuracy, where a prediction is considered correct if the reference segment produced by the DfT model is included in the model's top-k predictions. As shown in Figure 1, each LLM, although independent, performs similarly, with the fusion model consistently outperforming the average in all cases. Among all the LLM models, the fusion model achieves a Top-1 accuracy of 56% and a Top-2 accuracy of around 90%. This indicates that the model can capture the characteristics of the segments, while the observed accuracy loss suggests a trade-off between reduced information and performance, or that the segmentation is originally insufficiently distinct. These results also demonstrate that combining multiple LLMs provides a modest but consistent improvement in segment identification.

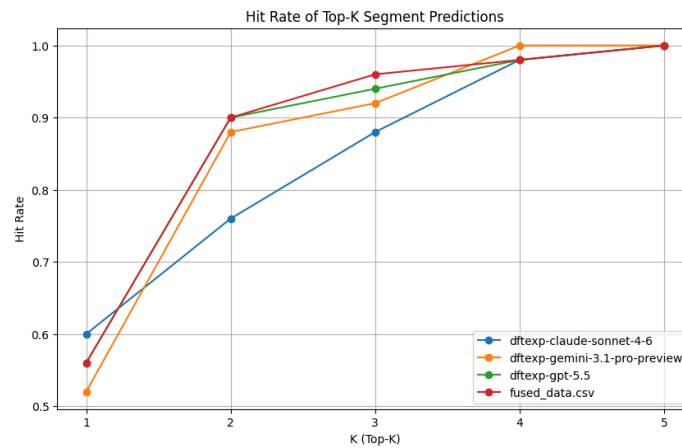


Figure 1. Top-K hit rate between LLMs' predictions and the original DfT segmentations

Future Work

We are currently refining the persona-labelling pipeline by testing different combinations of input attributes and assessing how these combinations affect the accuracy and stability of the outcome.

In the future, one possible direction would be integrating the proposed persona-labelling approach with the LLM-based AI Replanning agents in MATSim (Patwary et al., 2026). Once each synthetic agent is assigned a mobility persona, this behavioural profile can be used as contextual information in LLM prompts for selected replanning decisions based on specific mobility attitudes. This would address a key limitation of existing LLM-based replanning implementations, where agents may technically be able to query an LLM but still share the same generic behavioural context. By conditioning prompts on inferred personas, the LLM-based replanning mechanism could generate more differentiated and interpretable responses across agents. For example, a car-reliant agent, a low-income non-car-owning agent, and a flexible urban professional may respond differently to the same disruption, pricing intervention, or mobility-service improvement. In this way, persona-conditioned LLM replanning could move MATSim towards more behaviourally heterogeneous agent adaptation.

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